

UNIVERSAL NAVIGATION INTERFACE: Robot-Free Data for Wheeled Robot Navigation

<https://universal-nav.github.io/>

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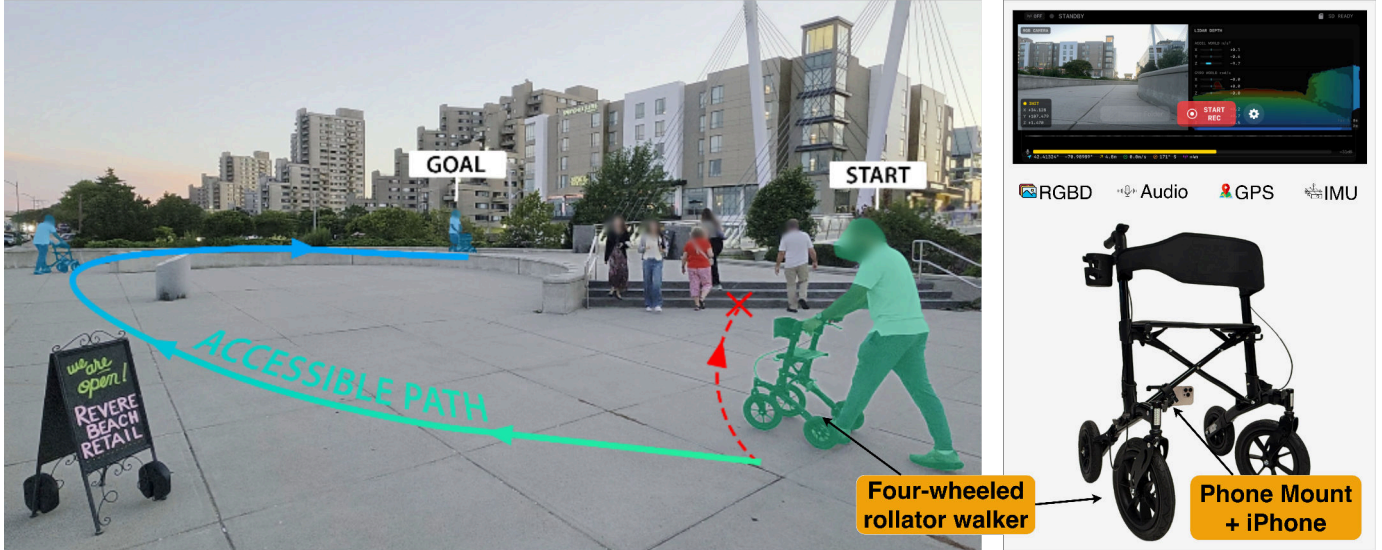


Fig. 1. At a route with stairs and an adjacent ramp, an unconstrained pedestrian may take the stairs (red path), while the rollator biases the collector toward the wheeled-feasible ramp. A mounted smartphone records synchronized RGB, depth, inertial, audio, and GPS data, which are post-processed into metric trajectories and observation-goal pairs for visual navigation.

Abstract—Collecting large-scale, task-relevant navigation data for wheeled mobile robots traditionally requires expensive, platform-specific expert teleoperation that is difficult to scale. To address this bottleneck, we present Universal Navigation Interface (UNI), a scalable, low-cost, and robot-free data collection paradigm for wheeled robot navigation. Our approach utilizes a commercially available four-wheeled rollator walker equipped with a standard smartphone mount and an iPhone. This setup requires no custom assembly, specialized engineering effort, or operator training. Crucially, the human-pushed walker acts as a physical feasibility filter: it cannot climb stairs, negotiate uncut curbs, or pass through narrow gaps,

naturally constraining demonstrations to wheeled-feasible routes. We demonstrate the efficacy of this robot-free, off-the-shelf data collection method through UNI, a diverse multimodal dataset of rollator-constrained urban pedestrian navigation trajectories with metric motion labels. We further evaluate this dataset with an existing visual navigation model. Fine-tuning on UNI improves metric trajectory accuracy and scale-invariant path shape, while exposing remaining bottlenecks in the coverage of rare execution states.

I. INTRODUCTION

The recent success of deep learning in autonomous navigation has been driven in large part by the availability of high-quality, and in-domain data [1], [2], [3]. Whether navigating urban sidewalks or unstructured off-road environments, modern learning-based navigation systems rely on large volumes of diverse trajectory data to generalize effectively [4], [5] or VLMs/foundational models [6], [7]. However, collecting data directly on robotic platforms can present a substantial operational bottleneck. Large-scale robot-native collection often requires sensor integration, hardware maintenance, power management, and sustained field operation, making it operationally demanding and difficult to scale.

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Existing navigation datasets present different trade-offs between scalability and embodiment fidelity. Robot-native datasets such as GoStanford, ViKiNG, GNM, and ViNT [8], [9], [10], [4] provide on-platform observations and robot-grounded motion trajectories. However, they still require deploying, maintaining, and operating robots, and their demonstrations reflect the embodiment and control interface of the source platform. Human-worn sensor datasets such as MuSoHu [11] avoid this robot-deployment bottleneck and capture rich interactions in public spaces, but introduce an embodiment mismatch through gait-induced motion and a sensor viewpoint typically higher than that of a mobile robot. Web-scale approaches such as CityWalker [5] further improve scalability, but do not impose wheeled-feasibility during collection. Unconstrained human and web video may therefore include stairs, uncut curbs, or narrow passages that are infeasible for wheeled platforms. Existing data sources thus offer strong robot grounding at the cost of physical platform deployment, or greater scalability without a collection-time physical constraint toward wheeled-feasible motion, motivating an interface that combines the advantages of both.

Manipulation faced the same bottleneck. UMI [12] decoupled data collection from the robot with a hand-held, embodiment-matched gripper, reusing an off-the-shelf policy rather than proposing a new architecture. We ask whether navigation admits the same decoupling, and introduce a low-overhead, robot-free collection paradigm using a commercial four-wheeled rollator walker (rollator) and a standard smartphone. The setup requires no custom hardware integration or specialized operator training. More importantly, the human-pushed walker acts as a physical feasibility filter: under normal operation, it cannot climb stairs, negotiate uncut curbs, or pass through gaps narrower than its footprint (Fig. 1). Unlike unconstrained walking or a wearable sensor rig, the rollator imposes a wheeled footprint during collection, so routes are filtered by the same constraint a mobile platform faces.

Modern smartphones provide a compact, commodity platform integrating cameras, inertial sensing, positioning, depth sensing on supported devices, and onboard compute. By combining these synchronized sensor streams with metric trajectory recovery, we convert each recording into observation-goal pairs with future motion expressed directly in meters. The resulting metric waypoint labels can be used directly by existing trajectory-prediction and visual-navigation architectures without collecting demonstrations on the target robot.

In summary, the contributions of this workshop paper are: (1) a low-overhead, robot-free collection paradigm in which a rollator acts as a physical feasibility filter, naturally constraining demonstrations toward wheeled-feasible routes; (2) A 37.2 km multimodal urban navigation dataset with metrically recovered trajectories, together with a depth-anchored scale recovery pipeline validated against established geometry and SLAM baselines; and (3) an evaluation with an existing visual navigation model showing that fine-tuning on UNI improves metric trajectory accuracy and scale-invariant path shape, while exposing execution scenarios that remain underrepre-

sented in existing navigation datasets.

II. SYSTEM OVERVIEW

In this section, we describe the complete UNI data collection rig, from collecting demonstrations traversable by wheeled platforms using the rollator and smartphone to post-processing the recorded sensor streams into synchronized, metrically scaled trajectories for visual navigation.

A. Rollator-Constrained Data Collection

UNI uses a commercially available rollator as both a sensor carrier and physical feasibility filter. Unlike unconstrained human walking, the rollator cannot climb stairs, mount uncut curbs, or pass through gaps narrower than its footprint; consequently, the operator is naturally constrained toward routes compatible with wheeled mobility. A smartphone is attached to the rollator using a standard phone mount, requiring no custom mechanical integration or platform-specific instrumentation (Fig. 1 right).

B. SensorVault: Data Collection App

For UNI data collection, we developed SENSORVAULT, an iOS application that records calibrated multimodal sensor streams on LiDAR-equipped iPhones (Fig. 1 right). The application supports any LiDAR-equipped iPhone; all the data was collected using an iPhone 16 Pro. The app records JPEG-compressed RGB images, ARKit LiDAR depth and confidence maps at 256×192 resolution, 100 Hz inertial measurements, GPS, audio, and 6-DoF visual-inertial (VIO) poses relative to the beginning of each recording. RGB and depth intrinsics are stored with each recording, and fixed focus is used to preserve calibration. All sensor streams share a monotonic clock and are stored in MCAP format, with low-confidence or out-of-range depth measurements marked invalid before storage.

Because recorded VIO can accumulate drift over time, navigation trajectories are recovered independently using the depth-anchored visual reconstruction pipeline described below. SENSORVAULT itself is not required by the UNI paradigm: for visual navigation, the core supervision consists of RGB observations paired with metrically scaled trajectories. This functionality can therefore be reproduced using conventional smartphone RGB video when paired with a suitable metric-trajectory recovery method.

C. Post Processing the Collected Data

To recover metric navigation trajectories, we process RGB recordings with Depth Anything 3 (DA3) [13] in 48-frame windows at 4 Hz (~ 12 s), with 12-frame (3 s) overlap. DA3 jointly predicts per-frame depth $\hat{D}_i$ and camera poses $T_i = [R_i \mid t_i]$, where $R_i \in SO(3)$ and $t_i \in \mathbb{R}^3$ denote the world-to-camera rotation and translation, respectively. As with monocular reconstruction, the predicted depth and camera translation within each window are defined only up to a common scale. Let $\hat{C}_i = -R_i^\top t_i$ denote the predicted camera center. For window w , the unknown metric quantities satisfy

$$D_i^m = s_w \hat{D}_i, \quad C_i^m = s_w \hat{C}_i, \quad (1)$$



Fig. 2. Representative scenes from the UNI dataset, illustrating the diversity of rollator-constrained urban pedestrian environments, including crowded sidewalks, building entrances, transit interfaces, wet pavement, and construction-constrained walkways.

where s_w is an unknown window-specific scale.

Consecutive windows are similarity-aligned over their overlapping frames and chained into a common relative coordinate frame. The estimated relative scale for each window is applied to both its camera centers and its predicted depths, yielding chained quantities $\tilde{C}_i$ and $\tilde{D}_i$ that share a single arbitrary unit across the segment. The RMS disagreement between overlapping pose estimates provides a join residual used to reject inconsistent joins.

We then resolve one absolute metric scale per chained segment using the smartphone’s co-registered LiDAR depth S_i , although the same procedure applies to any reconstruction method that provides dense depth and camera poses up to a common scale. Invalid, near-range and range-saturated LiDAR measurements are marked at capture time and excluded here. Let $\mathcal{F}$ denote the frames retaining sufficient valid depth support, and M_i the valid pixels of frame i after resampling S_i onto the predicted depth grid. We estimate a robust per-frame metric ratio and take its median across the segment,

$$\rho_i = \text{median}_{p \in M_i} \frac{S_i(p)}{\tilde{D}_i(p)}, \quad \hat{s} = \text{median}_{i \in \mathcal{F}} \rho_i. \quad (2)$$

Segments without sufficient metric support are rejected rather than assigned a forced scale; this occurs on 8.3% of candidate segments in the dataset. The metric trajectory is $C_i^m = \hat{s} \tilde{C}_i$. Future poses are expressed relative to the current observation over a horizon of 5 waypoints (1.33 s) and exported as metrically scaled observation–goal pairs for navigation-policy training.

III. THE UNI DATASET

The UNI dataset is an egocentric, rollator-constrained navigation corpus comprising 87 recording sessions collected over 17 days from June to August 2026 across Boston and New York City, spanning diverse urban environments, weather conditions, and times of day. Each recording contains synchronized RGB, LiDAR depth, inertial, audio, GPS, and visual-inertial odometry streams. The raw corpus contains 10.8 h of recordings, of which 8.0 h correspond to active walking, covering an estimated 37.2 km of travel. The remaining recordings capture non-walking periods, including transit by train or elevator, yielding to pedestrians, and waiting at crosswalks.

The UNI dataset spans 15 distinct surface types across urban, indoor, and natural environments: concrete sidewalks, brick and tactile paving, asphalt paths and roads, gravel, cobblestone, wet pavement, boardwalks, dirt, grass, sand, carpet,

tile, and terrazzo. Less common surfaces remain sparsely represented, motivating targeted future collection.

Although collected for wheeled-navigation research, the UNI dataset also captures naturalistic human mobility, including waiting at crosswalks, navigating around other pedestrians, and traversing ramps and curb cuts (Fig. 2). Because collection follows a pedestrian rather than a continuously driven platform, waiting arises as a natural consequence of the setting: 7.3% of exported training samples carry a near-zero motion label, against 0.38% for a mixture-weighted average over the datasets we could measure from ViNT’s training mix (54 of 80 hours used, weighted by ViNT’s reported hours; the unmeasured remainder is entirely continuous-motion off-road and on-road data, which would lower the mixture further).

We will release privacy-processed recordings in MCAP format, with faces and license plates blurred using EgoBlur [14]. The UNI dataset export also contains 1,087 trajectories and 117,512 frames sampled at approximately 4 Hz, with 793 training trajectories, 157 held-out clean trajectories, and 137 additional reserved evaluation trajectories that include wet-weather examples.

IV. EXPERIMENTS

We evaluate two components of UNI: the accuracy of the recovered metric trajectory labels and their utility for visual navigation. First, we validate the trajectory-labeling pipeline independently of the UNI dataset using TUM RGB-D [15] and KITTI [2]. Both benchmarks are divided into non-overlapping 8 s windows and processed at 4 Hz, matching the UNI pipeline. TUM provides dense indoor depth and motion-capture ground truth (65 windows), while KITTI provides an outdoor regime with substantially sparser and longer-range lidar depth (82 windows).

The learned geometry models recover trajectories for all TUM windows, whereas ORB-SLAM3 succeeds on 47/65 windows at 4 Hz, improving to 61/65 at the native recording rate. On successfully reconstructed windows, classical methods can match or outperform the learned models in trajectory shape; the advantage of the UNI pipeline is reliable short-window reconstruction together with metric scale recovery. Applying the same depth anchor to three different geometry backbones yields 3–5% median scale error. On KITTI, the anchor achieves 2.55% error with full lidar range and 5.92% with depth limited to 5 m, rejecting windows with insufficient depth support rather than returning an unsupported metric estimate.



Fig. 3. Qualitative example of an approach-to-wait event on held-out UNI dataset. The released ViNT model continues to predict forward motion during the wait, while the UNI-adapted model reduces its predicted path length toward zero. This example illustrates a behavior present in the data rather than an aggregate measure of wait-state performance.

We further test whether the UNI dataset provides useful supervision for learned visual navigation. We use ViNT [4] because it is an established goal-conditioned navigation model trained from observation–goal trajectory pairs, the same supervision format produced by UNI. This data interface is also representative of a broader class of learning-based navigation methods in which visual observations are paired with future motion targets. We therefore use ViNT as a testbed for whether robot-free UNI demonstrations can complement broad robot pretraining, rather than proposing a new navigation architecture.

TABLE I
ViNT EVALUATION ON THE HELD-OUT UNI TEST SPLIT.

Model	ADE (m) ↓	FDE (m) ↓	ADE _{shape} ↓	dir (°) ↓
Released ViNT	0.338	0.568	0.243	25.4
UNI scratch	0.305	0.509	0.233	25.4
UNI fine-tuned	0.249	0.412	0.193	22.8

Table I reports results on 67 held-out trajectories spanning 6 outings, disjoint from training at the outing level, over a 1.33 s horizon. Metric errors use the split’s measured spacing of 0.219 m/frame; ADE_{shape} refits one scalar per trajectory and is invariant to it, and heading error is measured at the final waypoint.

Fine-tuning on UNI reduces ADE from 0.338 m to 0.249 m and FDE from 0.568 m to 0.412 m, reductions of 26% and 28%, respectively, with a 21% improvement in scale-invariant path shape. Training from scratch on UNI alone yields substantially smaller reductions of approximately 10% in both ADE and FDE, indicating that UNI is most useful as complementary supervision for an existing navigation prior rather than as a replacement for broad robot pretraining. Under a one-sided sign test, FDE improves on all six outings ($p = 0.016$), while ADE improves on five of six ($p = 0.109$). Given the small number of outings, we treat these results as descriptive.

V. DISCUSSION AND FUTURE WORK

Our experiments suggest that UNI is most useful as a complement to broad robot pretraining rather than a replace-

ment for it. ViNT already provides a strong directional prior, while fine-tuning on UNI improves trajectory accuracy without sacrificing directional performance.

Rarer behaviors remain underrepresented. Waiting, yielding, and stop–resume transitions occur in the dataset, but are concentrated in a small number of independent events. Future collection will therefore focus on increasing event diversity, including matched blocked and clear traversals.

Future work will also explore separating path prediction from trajectory timing and improving geometric prediction on sharper turns. We further plan to evaluate transfer to a wheelchair and a quadruped using platform-specific low-level controllers.

VI. CONCLUSION

We introduced UNI, a low-overhead, robot-free interface for collecting wheeled-feasible navigation demonstrations using a commercial rollator and smartphone. The UNI dataset contains 37.2 km of multimodal urban navigation data with metrically recovered trajectories, obtained without deploying a robotic platform. Benchmark experiments validate the metric trajectory-recovery pipeline across multiple geometry backbones, while ViNT experiments show that UNI data complements broad robot pretraining by improving goal-progress and action-magnitude prediction while preserving directional accuracy. Together, these results support physically constrained human collection as a practical path toward scaling navigation supervision beyond robot-native data collection.

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